

Video 1: Conditional Probabilities, Clearly Explained!!!

Video Link: https://youtu.be/_IgyaD7vOOA

Channel: StatQuest with Josh Starmer

I. Core Concept: What is Conditional Probability?

- **Scenario:** A survey in "StatLand" (total 14 people) asking if they love **Candy** and/or **Soda**.
- **Contingency Table (Counts):**
 - Love Candy AND Love Soda: 2 people
 - Love Candy BUT NOT Soda: 4 people
 - Love Soda BUT NOT Candy: 5 people
 - Love NEITHER: 3 people
 - **Total People: 14**
- **Contingency Table (Probabilities):**
 - We can convert these counts to probabilities by dividing each by the total population (14).
 - $P(\text{Candy and Soda}) = 2/14 = 0.14$
 - $P(\text{Candy and No Soda}) = 4/14 = 0.29$
 - $P(\text{No Candy and Soda}) = 5/14 = 0.36$
 - $P(\text{No Candy and No Soda}) = 3/14 = 0.21$
- **Totals (Marginal Probabilities):**
 - Total who Love Soda = $2 + 5 = 7$ people. $P(\text{Love Soda}) = 7/14 = 0.5$
 - Total who Love Candy = $2 + 4 = 6$ people. $P(\text{Love Candy}) = 6/14 = 0.43$

II. Calculating Conditional Probability

- **The Question:** What is the probability that someone loves soda, **GIVEN THAT** we know they love candy?
- **Notation:**

$$P(\text{Love Soda} \mid \text{Love Candy})$$

- The vertical bar $\mid$ means "given that".
- **The Intuition (Shrinking the Universe):**
 - "Given that we know they love candy" means we are **no longer looking at all 14 people**.
 - We restrict our focus **only** to the 6 people who love candy. This is our new "universe" or denominator.
 - **Out of these 6 people**, how many *also* love soda?
 - Looking at the table, 2 people love *both*. This is our numerator.
- **Calculation (Using Counts):**

$$P(\text{Love Soda} \mid \text{Love Candy}) = \frac{\text{Count}(\text{Love Soda AND Love Candy})}{\text{Count}(\text{Love Candy})} = \frac{2}{6} = 0.33$$

- **Deriving the General Formula:**
 - We can divide both the numerator and the denominator by the *total population* (14) without changing the result.

$$P(\text{Love Soda} \mid \text{Love Candy}) = \frac{2/14}{6/14}$$

- Notice what these terms are:
 - $2 / 14$ is the original, unconditional probability **ParseError: KaTeX parse error: Unexpected end of input in a macro argument, expected '}' at end of input: ...AND Love Candy**).
 - $6 / 14$ is the original, unconditional probability **ParseError: KaTeX parse error: Unexpected end of input in a macro argument, expected '}' at end of input: ...ext{Love Candy}**).
- **The General Formula for Conditional Probability:**

$$P(A | B) = \frac{P(A \text{ AND } B)}{P(B)}$$

- Where:
 - $P(A | B)$ is the probability of A *given* B.
 - $P(A \text{ AND } B)$ is the probability of *both* A and B happening (the intersection).
 - $P(B)$ is the probability of B happening (the "given" part, our new denominator).

III. The Multiplication Rule (Rearranging the Formula)

- If we take the main formula:

$$P(A | B) = \frac{P(A \text{ AND } B)}{P(B)}$$

- And multiply both sides by $P(B)$, we get the **Multiplication Rule**:

$$P(A \text{ AND } B) = P(A | B) \times P(B)$$

- This is very useful. It says the probability of *both* things happening is the probability of the "given" thing happening ($P(B)$), *multiplied by* the probability of the other thing happening *after* the first one is already true ($P(A | B)$).

Video 2: Bayes' Theorem, Clearly Explained!!!!

Video Link: <https://youtu.be/9wCnvr7Xw4E>

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I. Introduction & Review of Conditional Probability

- **Prerequisite:** This video assumes you understand **Conditional Probability** (from the first video).
- **Review:** Using the same "StatLand" soda/candy example, the video reviews two conditional probabilities:
 - $P(\text{No Candy} | \text{Love Soda})$
 - This asks: *Given that* a person loves soda (our universe is the 7 soda lovers), what's the chance they *don't* love candy?
 - Calculation: $\frac{P(\text{No Candy and Love Soda})}{P(\text{Love Soda})} = \frac{5/14}{7/14} = \frac{5}{7} = 0.71$
 - $P(\text{Love Soda} | \text{No Candy})$
 - This asks: *Given that* a person does *not* love candy (our universe is the 8 non-candy lovers), what's the chance they *love* soda?
 - Calculation: $\frac{P(\text{No Candy and Love Soda})}{P(\text{No Candy})} = \frac{5/14}{8/14} = \frac{5}{8} = 0.63$

- **Key Observation:** In both cases, the event we care about is the *same* (the 5 people who love soda but not candy). The **numerators are identical**. The only difference is what we *know*, which changes the denominator.

II. Deriving Bayes' Theorem Algebraically

- **The Problem:** What if we don't know the joint probability `ParseError: KaTeX parse error: Unexpected end of input in a macro argument, expected '}' at end of input: ... and Love Soda`? Can we still solve for the conditional probabilities?
- **Step 1:** Start with the two multiplication rules derived from the probabilities above.
 - $P(\text{No Candy and Love Soda}) = P(\text{No Candy} \mid \text{Love Soda}) \times P(\text{Love Soda})$
 - $P(\text{No Candy and Love Soda}) = P(\text{Love Soda} \mid \text{No Candy}) \times P(\text{No Candy})$
- **Step 2:** Since both equations equal the *same thing*, we can set them equal to each other:

$$P(\text{No Candy} \mid \text{Love Soda}) \times P(\text{Love Soda}) = P(\text{Love Soda} \mid \text{No Candy}) \times P(\text{No Candy})$$

- **Step 3:** Now, we can solve for whichever conditional probability we want. Let's solve for $P(\text{Love Soda} \mid \text{No Candy})$ by dividing both sides by $P(\text{No Candy})$.
- **The Result (Bayes' Theorem):**

$$P(\text{Love Soda} \mid \text{No Candy}) = \frac{P(\text{No Candy} \mid \text{Love Soda}) \times P(\text{Love Soda})}{P(\text{No Candy})}$$

This is **Bayes' Theorem**. It lets us "flip" the conditional probability.

III. The Standard Formula & Why It's a "Big Deal"

- If we use generic A and B for the events:

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

- **Terminology:**
 - $P(A \mid B)$ (**Posterior**): The probability we want to find (the "updated" probability of A *after* knowing B).
 - $P(B \mid A)$ (**Likelihood**): The probability of B *given* A (the "flipped" probability, which is often easier to know).
 - $P(A)$ (**Prior**): The original, unconditional probability of A (our "prior" belief before knowing B).
 - $P(B)$ (**Marginal Likelihood**): The original, unconditional probability of B .
- **Why is it useful?**
 - When we have all the data in a contingency table, Bayes' Theorem isn't necessary.
 - **Its power comes from when we *don't* have all the data.**
 - In the real world, we often know $P(B \mid A)$ but *want* to know $P(A \mid B)$.
 - **Classic Example:**
 - We know the probability of having a **Symptom** *given* you have a **Disease**. $P(\text{Symptom} \mid \text{Disease})$. This comes from clinical studies.
 - A patient comes to you with the **Symptom**. You *want* to know the probability they have the **Disease**. $P(\text{Disease} \mid \text{Symptom})$.
 - Bayes' Theorem lets us use the $P(\text{Symptom} \mid \text{Disease})$ that we know, along with our "prior" belief about how common the disease is ($P(\text{Disease})$), to find the $P(\text{Disease} \mid \text{Symptom})$ that we need.

- This ability to update our beliefs (priors) with new evidence (likelihoods) to get a new, refined belief (posterior) is the foundation of **Bayesian Statistics**.